Java Loops - Course Notes
DevTutor evaluation document, page 1 of 3

1. What is a loop?
A loop repeats a block of code several times.
Loops save you from writing the same line again and again.

2. The for loop
Use a for loop when you know how many times the code must repeat.
A for loop has three parts: a start value, a condition and an update.
The loop runs while the condition is true.

3. The enhanced for loop
The enhanced for loop goes through every element of an array or a list:
for (String name : names) { System.out.println(name); }
It is the simplest choice when you do not need the position of the element.

4. The while loop
Use a while loop when you do not know in advance how many times to repeat.
The condition is checked before each repetition:
while (count < 10) { count++; }

5. The do-while loop
A do-while loop always runs its body at least once,
because the condition is checked after each repetition.

6. break and continue
break stops the loop immediately.
continue skips the rest of the current repetition and starts the next one.

7. Infinite loops
If the condition never becomes false, the loop never ends.
Always make sure something inside the loop moves it toward the end.

8. DevTutor course rules for loops

In this course, every loop must follow these two naming rules.

Rule 1: Always name loop counters step, never i.
Reason: step reads like one step of the loop and is clearer for beginners.

Rule 2: Use limit for the maximum number of loop iterations.
Reason: limit makes the maximum visible at a glance.

Exercises that ignore these rules lose points in this course.